

BARNSTABLE, MA, USA

WMO: 725067

Lat: 41.669N

Lon: 70.280W

Elev: 55

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 94720

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	14.5	-6.6	3.9	12.9	-1.6	5.1	18.1	29.4	32.4	26.4	35.5	9.2	300	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	84.6	73.2	81.9	71.8	79.6	70.6	76.0	81.5	74.6	79.2	73.3	77.3	11.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.0	127.2	78.8	73.0	122.5	77.5	72.1	118.8	76.5	39.3	81.4	38.1	79.4	36.9	77.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.2	19.1	3.1	90.2	5.5	3.8	-0.8	92.9	-4.1	95.1	-7.2	97.2	-11.2	100.0		
			1.8	78.4	5.4	1.7	-2.1	79.6	-5.3	80.7	-8.3	81.6	-12.2	82.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	31.0	32.1	37.3	46.1	55.7	64.7	71.7	70.7	63.9	53.7	44.7	36.6
	DBStd	15.85	9.30	7.92	7.45	6.20	6.37	6.30	5.00	5.24	6.28	7.29	7.72	8.20
	HDD50	2315	590	502	398	147	16	0	0	0	0	44	196	422
	HDD65	5739	1055	922	859	566	295	81	7	11	95	356	610	882
	CDD50	2606	0	0	4	31	194	442	674	643	419	159	35	5
	CDD65	556	0	0	0	1	8	73	216	189	63	6	0	0
	CDH74	2721	0	0	0	9	50	313	1181	977	180	11	0	0
	CDH80	490	0	0	0	2	10	63	242	155	17	1	0	0
Wind	WSAvg	9.1	9.8	9.8	10.5	9.9	9.2	8.8	8.6	8.0	8.0	8.8	9.2	9.2
Precipitation	PrecAvg	43.8	3.6	3.6	3.5	3.9	3.7	3.2	3.0	3.4	3.4	3.7	4.5	4.2
	PrecMax	61.9	7.6	9.1	9.5	7.8	9.1	7.7	10.0	10.2	10.4	8.3	8.7	9.2
	PrecMin	27.5	1.2	1.0	0.9	1.0	0.6	0.2	0.6	0.2	0.2	1.4	0.6	0.6
	PrecStd	7.7	1.9	1.9	1.9	1.9	2.1	2.1	2.2	2.3	2.1	1.8	2.0	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	54.3	61.6	70.2	80.8	86.4	89.7	87.6	82.1	75.2	64.6	59.7
		MCWB	51.8	49.7	51.6	59.0	66.1	72.6	75.9	75.3	71.9	68.0	59.9	56.2
	2%	DB	50.5	50.1	54.6	62.6	73.2	81.1	85.4	83.7	78.5	71.0	62.2	55.6
		MCWB	48.5	47.0	48.1	53.5	62.9	69.6	73.6	73.6	70.0	65.5	58.5	53.2
	5%	DB	47.7	46.9	51.6	58.8	69.6	77.3	82.3	81.4	75.5	68.0	59.7	53.2
		MCWB	45.2	43.4	47.2	51.5	60.4	67.4	71.9	72.4	69.1	63.7	56.0	51.1
	10%	DB	44.9	44.6	48.2	55.8	65.8	74.4	80.0	79.2	73.4	65.2	57.0	49.9
		MCWB	42.2	41.2	44.4	49.7	58.4	66.2	70.9	71.2	68.3	61.1	53.6	47.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	51.9	54.4	60.2	68.5	74.5	78.0	78.0	74.9	69.9	62.2	57.4
		MCDB	52.8	52.9	57.4	66.4	76.4	82.3	87.1	84.9	79.3	72.6	63.8	58.9
	2%	WB	49.4	47.7	50.7	56.4	64.9	72.1	76.0	76.3	73.0	66.9	59.6	54.1
		MCDB	50.4	49.3	53.1	60.4	70.9	78.2	82.4	81.2	76.1	69.6	61.8	55.2
	5%	WB	46.0	44.6	47.9	53.9	62.5	70.2	74.6	75.0	71.3	64.5	57.0	51.3
		MCDB	47.2	46.4	50.7	56.7	67.0	74.9	79.6	79.0	73.9	66.9	59.1	52.6
	10%	WB	42.5	41.4	45.2	51.5	60.2	68.3	73.3	73.6	69.8	62.3	54.3	47.7
		MCDB	44.5	44.1	47.4	54.5	64.6	72.7	77.4	77.3	72.6	64.7	56.4	49.7
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.6	13.9	14.7	14.5	14.0	13.9	14.2	15.2	15.5	14.6	14.3
		MCDBR	15.8	17.3	17.0	18.9	19.3	17.4	15.9	14.7	14.8	16.1	14.9	15.9
		MCWBR	15.4	15.1	13.5	12.1	10.9	9.2	8.0	7.7	9.6	12.0	13.5	16.0
	5% WB	MCDBR	15.5	16.9	15.7	15.2	16.8	14.7	13.2	12.3	12.1	14.4	13.2	15.6
		MCWBR	16.1	16.3	14.2	11.8	10.5	8.9	7.3	7.2	9.2	12.4	13.4	16.6
Clear-Sky Solar Irradiance	taub	0.310	0.315	0.336	0.375	0.409	0.461	0.484	0.440	0.374	0.347	0.319	0.307	
	taud	2.516	2.503	2.480	2.394	2.342	2.230	2.175	2.292	2.456	2.532	2.567	2.565	
	Ebn at Noon	268	285	290	284	275	260	252	261	273	269	263	260	
	Edh at Noon	23	27	31	36	39	44	46	40	31	26	22	20	
All-Sky Solar Radiation	RadAvg	548	835	1165	1511	1706	1831	1891	1702	1383	940	625	466	
	RadStd	43	79	90	161	165	167	104	89	97	79	44	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171	+81

Nomenclature: See separate page